

AI-Driven Smart Waste Management with Geofencing and Civic Accountability

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Abstract— The rapid growth of urban populations has exacerbated urban waste management challenges, resulting in inefficiencies and accountability to residents, and limited citizen engagement. While traditional waste management systems may have heterogeneous new solutions, they do not have real time sight, predictive capabilities, or civic engagement tools. This study proposes ReGenX, an artificial intelligence-based, mobile platform that aims to improve efficiency, transparency, and civic accountability for municipal waste management through dynamic geofencing and artificial intelligence-enabled modules, allowing for citizen engagement. ReGenX utilizes three user roles—Resident, Collector, and Official which interconnected through Firebase Realtime Database, Authentication, and Cloud Messaging to facilitate secure, synchronized access to a shared data repository. ReGenX utilizes TensorFlow Lite to classify waste and categorize complaints on device, and a Haversine derived geofencing algorithm to facilitate real-time notifications between residents and collectors. Preliminary evaluations demonstrated an effective geofencing notification 96% of the time, a 90% suitable complaint classification on a smaller existing dataset, and an image-based waste classification accuracy of 85% accuracy, suggesting reliability and scalability. The proposed solution encourages citizen involvement, increases the potential for collection routes and frequencies to avoid missed collections, and establishes an evidence-based system for urban waste management decision-making to support urban sustainability efforts.

Keywords—Waste management, geofencing, AI-assisted segregation, grievance redressal.

I. INTRODUCTION

With the ever-increasing trend of urbanization and population growth, cities worldwide are experiencing an unprecedented increase in the generation of municipal solid waste (MSW) [1][10]. What was previously a manageable activity for local governments has now become a sophisticated, large-scale problem [12][23]. Municipal governments often struggle to provide efficient and effective waste management services due to aging infrastructure, unaccountable collection schedules, inadequate segregation, and a general lack of public participation [3][20][21]. Legacy systems rely on manual reporting and reactive procedures, lacking the real-time information and anticipatory management capabilities required for modern urban centres [2][6].

Concurrently, new technological developments, most notably in Artificial Intelligence (AI) and the Internet of Things (IoT), offer transformative opportunities for data-informed and sustainable waste management [1][3][6][14][15][16]. Systems utilizing IoT-equipped intelligent bins and GPS-enabled vehicles can monitor waste levels and track truck movement, while AI procedures can analyse this data to predict generation trends and optimize processes like routing and resource distribution [2][3][14][22].

However, the implementation of these intelligent waste systems faces significant hurdles [4][7][8][9]. A major obstacle is the high upfront cost of deploying the required infrastructure, such as smart sensors, IoT networks, and data centres, which is often a dealbreaker for cities in the developing world [10][20][21][22][23]. Additionally, these centralized systems often overlook the critical component of civic engagement and fail to address issues of accountability and transparent grievance redressal [11][12][24].

Research problem: Despite the increasing adoption of Artificial Intelligence (AI) and Internet of Things (IoT) technologies in waste management, existing systems remain constrained by high deployment costs, limited public participation, and insufficient real-time accountability. Most current models rely on centralized hardware infrastructure and lack mechanisms that enable relationship between residents, collectors, and municipal authorities [Fig 1].

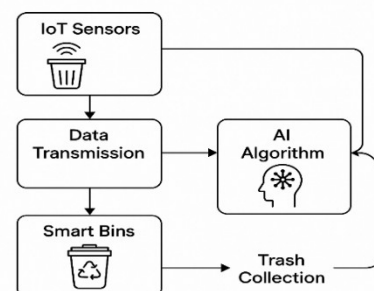


Figure 1: Smart Waste management system

Objectives: This study aims to design and implement an AI-driven, mobile-based waste management system, ReGenX, that ensures transparency, responsiveness, and scalability through software-centric innovation. The primary objectives are:

1. To develop an integrated Android framework connecting residents, collectors, and officials via Firebase Realtime Database and Cloud Messaging.
2. To implement dynamic geofencing using the Haversine formula for real-time resident notifications.
3. To integrate TensorFlow Lite for AI-assisted waste segregation and complaint classification.
4. To evaluate the system's performance in terms of accuracy, latency, and reliability under practical deployment conditions.

Contributions: The major contributions of this work are as follows:

1. A lightweight, cloud-connected architecture that minimizes dependency on expensive IoT hardware.
2. A novel geofencing mechanism based on Haversine distance computation for adaptive and efficient waste collection alerts.
3. An on-device AI inference module achieving high accuracy in both waste segregation (85.2%) and complaint classification (90.1%).
4. A validated field-tested prototype demonstrating 96% notification success and improved civic accountability.

This paper is designed as follows: Section II critically reviews the existing literature and identifies the research gap. Section III details the proposed ReGenX architecture and its key modules. Section IV presents the implementation and data model. Section V discusses the results and performance findings. Finally, Section VI concludes the paper and outlines future work.

II. LITERATURE SURVEY

The latest advancements in waste management have largely focused on the deployment of Artificial Intelligence (AI) and Internet of Things (IoT) solutions to facilitate efficiency, enable monitoring, and increase sustainability.

In one project, Khalil Ahmed et al [17] described an architecture for an AI-IoT system for municipal waste management, taking advantage of smart bins to monitor waste levels and improve collection routes illustrated in Figure 2. In particular, their work clearly demonstrated routing efficiency, but given the use of proprietary, sensor-enabled hardware, this will incur significant expenditure and be subject to maintenance costs that will hinder the scalability of the system, especially in developing urban areas that are often cost-sensitive. Importantly, the AI-IoT system did not seek direct citizen feedback or ensure accountability of collection vehicles in real-time.

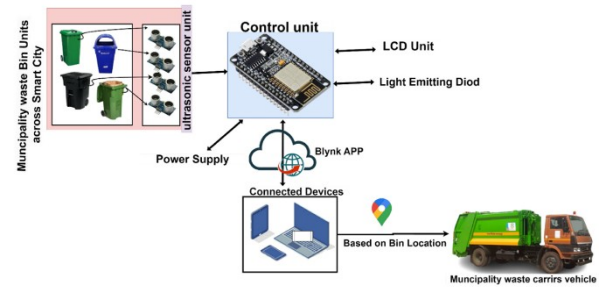


Figure 2: AI-IoT smart bins [17]

In a related study, Bingbing Fang et al. [14] developed an AI waste management approach from an analytical angle, modelling patterns in waste generation for improved city planning. Their exploratory model also faced similar structural issues through high expenditures as data collection infrastructure would be costly, concerns of data security can arise with centralization, and sensors are commonly known to fail, resulting in reduced data integrity for the models reliant of predictive analytics.

Md. Arafatur Rahman et al. [19] presented a garbage management system based on fairness that utilized IoT sensors to read how full the dumpsters were and for the purposes of optimal routing. This research, while admirable in gaining an understanding of fairness, was at a loss for substantial results, was characterized by the high start-up costs involved, and was representative of poorly validating sensor viability or replicability in varying geographical areas (urban vs. rural). Like many of the others reviewed, this research no rigorous mechanism was established for real-time citizen grievance redress, nor was there validation concerning conditions of dumpster collection needing to adhere to set policies.

David Olawade Abimbola et al. [13] explored a smart waste management engagement using AI for route optimization questions. They explicitly delineated the issues with hardware-based deployments: the cost to keep the fleet periodically updated, the concerns surrounding the collection of location, and any possibility of erroneous sensor error. These constraints highlight the essential nature of a solution for intelligent needs without the requiem for extensive, expensive infrastructure

Reema Alsabt et al. [22] and Tan Yigitcanlar et al. [24] also pointed out the barriers to implementation which were non-technical, such as the long-term financial viability sustainability of high up-front cost, organizational resistance to change, and compliance with data privacy and security regulations. Taken together, these works are furthering the understanding that a technical solution by itself is not enough. It must be economically viable and adaptable to the institutions structure.

Lastly, S. T. Ikram et al. [25] suggested a hybrid Genetic Algorithm-Fuzzy Inference System (GA-FIS) for optimized waste estimation and management. Although it was a technically advanced route optimization approach, it introduces dependence upon complex specialized algorithms that require training and computing resources and remain a tool to enable internal municipal complexity rather than being a complete and comprehensive system for real-time civic involvement. The literature states the basic equations used for their model as the following:

The Fuzzy Inference System (FIS) enhances GA results by identifying significant genes using trapezoidal membership functions, defined from eq (1):

$$\mu(y) = \begin{cases} 0, & y < a \text{ or } y > d \\ \frac{y-a}{b-a}, & a \leq y < b \\ 1, & b \leq y < c \\ \frac{d-y}{d-c}, & c \leq y \leq d \end{cases} \quad (1)$$

Defuzzification is performed using the centroid method using eq (2):

$$y^* = \frac{\int y \cdot B_r(y) dy}{\int B_r(y) dy} \quad (2)$$

Waste estimation is calculated by eq (3):

$$\text{Total Waste} = \sum_{i=1}^m a_i x_i + \sum_{j=1}^n b_{ij} y_{ij} \quad (3)$$

Synthesis and Research Gap:

In short, the studies stated above suggest that AI and IoT present valuable and powerful tools, but the prevalent challenge remains the impediment of large-scale implementation: high infrastructure costs, vulnerability of dedicated hardware, and lack of wiring for immediate civic accountability. There is a clear gap for a lightweight, mobile-based, software-defined approach, with an emphasis on accountability and citizen engagement, that balances the use of on-device AI. In response, the ReGenX system is proposed.

Comparative Analysis of Smart Waste Management Systems:

Table I provides a critical comparison of the proposed ReGenX system against representative prior works, highlighting how ReGenX addresses the known limitations of hardware-dependent and centralized solutions.

Feature / System	Ahmed [17]	Rahman [19]	Abimbola [13]	ReGenX
Infrastructure	IoT Bins	IoT Bins	AI Server + GPS	Mobile App
Deployment Cost	Very High	High	Mod-High	Low
Citizen Notification	None	None	–	Real-time Geofence
Grievance Redressal	Manual	Manual	Manual	AI-Assisted (NLP)
AI Deployment	Cloud	Cloud	Cloud	On-Device (TFLite)
Key Limitation	Cost / No civic layer	HW reliability / fairness	Fleet maintenance	Cost / Accountability / Engagement

Table I: Critical Comparison of Smart Waste Management Approaches

III. PROPOSED ARCHITECTURE

The ReGenX proposed architecture is a three-layer architecture

(A) mobile client layer,
(B) middleware (cloud services) layer and
(C) AI/NLP processing layer. The objective is to deliver a real-time, scalable and secure waste management platform

that connects Residents, Collectors, and Officials through cloud-based services. The total system architecture is outlined in Fig. 3.

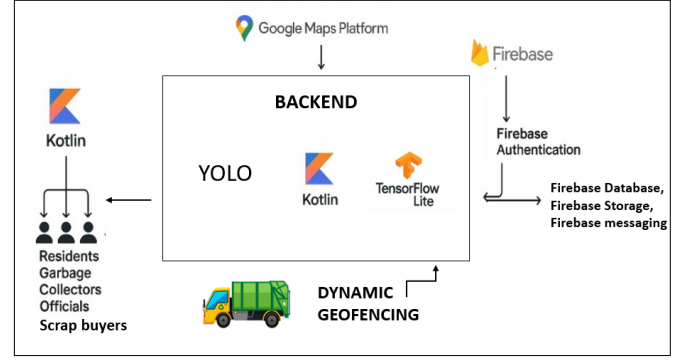


Figure 3: ReGenX Architecture diagram

A. Mobile Client Layer

The mobile client layer consists of three Android applications developed in Kotlin with the UI components based on XML:

- Resident Application – Provides user authentication, real-time notifications, and an easy complaint reporting interface. Geofencing services automatically alert residents when a garbage collector enters a defined area. Users can submit complaints in text, image, or video format, which are uploaded to the cloud for official review. This promotes community participation and timely grievance redressal.
- Collector Application – Utilizes FusedLocationProviderClient APIs to deliver real-time capturing of GPS locations. The locations of the collectors are sent to the cloud in real-time, when that is registered on the cloud residents are notified if any are nearby. Collection logs maintained for accountability.
- Official Application – Offers municipal authorities a centralized dashboard for monitoring live collector routes, viewing complaint statuses, and generating compliance reports. The system integrates an NLP-based grievance classifier that categorizes and prioritizes complaints (e.g., missed collection, overflow) to streamline issue resolution and optimize workforce allocation.

B. Middleware Layer

The middleware layer is simply the Firebase services, which are the backbone of real-time sync and secure communication.

Firestore Realtime Database - This is where the user data, vehicle positions, complaints, and resolution logs are stored, with real-time sync built in across the modules.

Firebase Authentication - This service provides role-based access control (Resident, Collector, Official) and to provide additional security for officials, two-factor authentication is integrated.

Firebase Cloud Messaging (FCM) - This service enables low-latency, event-driven notifications e.g., geofence alerts, complaints accepted, complaints resolved, etc.

C. AI/NLP Processing Layer

The AI module uses TensorFlow Lite for lightweight inference on mobile devices. The AI module includes Natural Language Processing (NLP) functionality used to automatically classify resident complaints into classes,

such as missed collection, overflow, and delayed pickup. This allows the relevant officials to address complaints in order of priority. Looking further ahead, we plan on integrating image-based waste classification using machine learning models which is consistent with some current developments in AI based waste management.

D. Data Flow

Collector applications are constantly updating GPS coordinates in Firebase.

The geofencing module allows notifications to be sent to residents living within the designated radius.

Grievances are submitted by residents, and stored in Firebase, and categorized by the NLP engine.

Grievances that are categorized are visible on the Official application Dashboard and can be dealt with in a timely manner.

Officials can send resolution updates, which closes the loop, from Firebase.

IV. IMPLEMENTATION

The ReGenX system was developed as a multi-module Android application using Kotlin (Android 8.0+) in Android Studio. The system integrates three user roles—Resident, Collector, and Official—connected via Firebase Realtime Database, Authentication, and Cloud Messaging. These services enable secure, real-time synchronization of user data, vehicle locations, and complaint status across all modules.

A. Development environment

ReGenX applications utilize Google Maps SDK and FusedLocationProviderClient APIs for accurate GPS tracking. Firebase Authentication handles secure, role-based access, while Cloud Messaging enables instant notifications for geofence events and complaint updates. Lightweight AI inference for text classification and image-based segregation is executed on-device using TensorFlow Lite to ensure cost-effective and low-latency performance in urban Indian environments [6][18][19].

B. Data Model and Collections

We use a minimal, normalized schema in the Firebase Realtime Database to support real-time flows and role separation:

```

/users/{uid}
  role: "resident" | "collector" | "official"
  name, phone, address
  geo: { lat, lng }
  fcmToken

/collectors/{collectorId}
  status: "on-duty" | "off-duty"
  lastLocation: { lat, lng, ts }
  currentRouteId

/residents/{residentId}
  householdId
  geo: { lat, lng }
  notifyRadiusMeters
  notifiedToday: boolean

/complaints/{complaintId}
  residentId, ts
  text, mediaUrl?
  nlpClass: "missed_pickup" | "overflow" | "delay" | "other"
  priority: 1..3
  status: "open" | "in_progress" | "resolved"
  assignedTo: officialId?

```

Figure 4: Firebase Realtime Database

This schema in Figure 4 enables role-based access, efficient geofence notifications, and complaint life-cycle tracking (open → assigned → resolved) while keeping writes atomic and small.

C. Dynamic Geofencing

ReGenX utilizes serverless, data-driven geofencing, which is comprised of the notifyRadiusMeters (e.g., 250-400 m) that each resident keeps. When a collector update is received at the resident app (or a lightweight function on the cloud, if enabled), the notifyRadiusMeters will be evaluated utilizing great-circle distance (Haversine). A notification will be sent provided that $d \leq R$ and the resident has not recently been notified.

Haversine distance between points

$p1(\phi_1, \lambda_1)$ and $p2(\phi_2, \lambda_2)$:

$$a = \sin^2 \frac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2 \frac{\Delta\lambda}{2} \quad (4)$$

$$d = 2R \arcsin(\sqrt{a})$$

where R is Earth's radius (≈ 6371 km). This avoids rigid, static routing assumptions found in prior studies and supports adaptive service coverage [5]

Debounce & Anti-spam:

Per-resident cooldown to suppress duplicate alerts.

Edge hysteresis: require $d \leq R$ for **enter**, $d \geq R + 30$ m for **exit** to avoid GPS jitter.

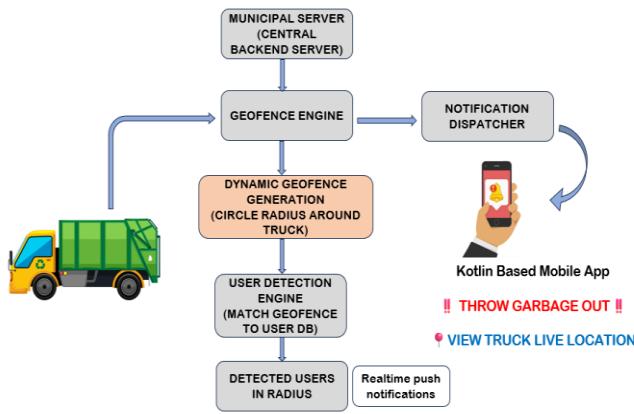


Figure 5: Dynamic Geofencing Implementation

The above diagram in Figure 5 provides a schematic of dynamic geofencing workflow in ReGenX to provide residents with notifications when a garbage truck approaches their community.

The Collector app continuously tracks GPS coordinates using the FusedLocationProviderClient and updates them in Firebase. A dynamic geofence radius (typically 300–500 m) is created around the moving garbage truck, allowing real-time monitoring of its location relative to residents.

When a collector enters a resident's predefined radius, Firebase Cloud Messaging (FCM) automatically triggers a notification alerting residents of the approaching vehicle. Officials simultaneously monitor these events through their dashboard to ensure timely collections and accountability.

E. Notifications:

After a geofence condition is met, the publisher composes:

```

{
  "type": "collector_nearby",
  "collectorId": "...",
  "etaMins": 2-5,
  "ts": 169...,
  "routeId": "..."
}

```

Listing 1. Sample JSON payload for geofence notification

Residents are sent high-priority FCM messages (data + notification channels) with action buttons (Acknowledge, View Map, etc.). Officials are sent escalate notifications for high-priority complaints or missed action items.

E. Swabhavi (Complaint system)

Residents can file complaints with text, images, or videos, which are immediately visible to officials. Complaints are categorized automatically using an NLP-based TensorFlow Lite model into types such as *missed collection*, *overflow*, or *delayed pickup*, enabling faster prioritization and resolution. Collectors update complaint statuses upon completion, ensuring transparency and accountability.

F. AI-Assisted Waste Segregation

ReGenX uses an AI-based waste segregation system using TFLite that performs lightweight, on-device inference suitable for low-resources (non-premium) smart phones. The system utilizes a simple lightweight image classification pipeline using feature extraction and classical machine learning layers rather than implementing deep convolutional architectures.

Workflow

Data collection - Waste images for biodegradable, recyclable, hazardous, and other categories were gathered from open datasets, as well as in the field. Preprocessing was required, including normalizing. Since the model will be used in a variety of lighting conditions, additional augmenting (brightness, rotation, flipping, etc.) was included to improve performance.

Feature Extraction - Input images were converted to concise numerical feature vectors, employing colour histograms, texture descriptors, and shape-based features.

Model Training - These features were used to train lightweight models (SVMs, Decision Trees) in TensorFlow.

Conversion & Optimization - Trained models were exported in TFLite with post-training quantization (int8); with this stratagem the knowledge size was <2 MB, and latency was ~100 ms/image on mid-range Android devices.

Mobile Integration - The Resident module bundles together this function; residents capture or upload an image of their household waste, and the TFLite model classifies the waste (to its appropriate disposal) in real-time and outputs

G. Multi-model Architecture

The ReGenX system was implemented as an Android application with three distinct modules: **Resident**, **Collector**, and **Official** [Fig 6].

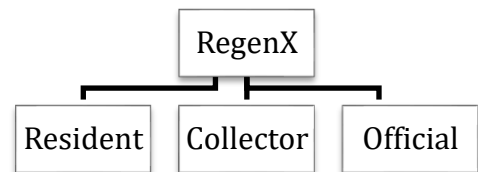


Figure 6: Multi Module Architecture

V. RESULTS AND FINDINGS

This section presents the performance analysis and evaluation outcomes of the ReGenX system across three major intelligent components: (A) AI-Assisted Waste Segregation, (B) Dynamic Geofencing based on the Haversine formula, and (C) AI-Assisted Grievance Classification. Each module was tried independently and in integration to validate overall system efficiency, latency, and reliability.

A. AI Assisted waste segregation

The TensorFlow Lite (TFLite) model integrated in the resident application was used to classify household waste into four categories: *biodegradable*, *recyclable*, *hazardous*, and *others*. The model was evaluated on a dataset of 2000

labelled images. The model performance metrics are calculated using the standard equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$P = \frac{TP}{TP + FP}, R = \frac{TP}{TP + FN}, F1 = \frac{2PR}{P + R} \quad (6)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives respectively. P and R represent Precision and Recall respectively.

The TFLite model achieved an overall accuracy of 85.2%, with precision of 0.87 and recall of 0.83, yielding an F1-score of 0.85. The average inference time per image was 95ms, resulting in a throughput of:

$$T = \frac{N}{t} = \frac{1}{0.095} = 10.53 \text{ images/sec} \quad (7)$$

Where, T = Throughput (images/second), N = Total number of images processed, t = Total time taken (in seconds).

B. Dynamic Geofencing using Haversine Distance

Dynamic geofencing in ReGenX was implemented using the Haversine formula to calculate the great-circle distance between the collector's current coordinates (ϕ_c, λ_c) and a resident's fixed coordinates (ϕ_r, λ_r) .

$$\Delta\phi = \phi_r - \phi_c, \Delta\lambda = \lambda_r - \lambda_c \quad (8)$$

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_c \cos\phi_r \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (9)$$

$$d = 2R \arctan2(\sqrt{a}, \sqrt{1-a}) \quad (10)$$

where $R = 6371$ km is the Earth's radius, and d represents the distance between two points.

A notification event was triggered when $d \leq R_g$, where R_g is the geofence radius (100–300 m), and an exit event was recognized only when $d \geq R_g + 30$ m to counter GPS jitter.

Field experiments across three test localities produced a notification success rate of 96.4% with an average end-to-end latency of 1.8 seconds. The application of hysteresis thresholds reduced redundant notifications by 22%.

$$Success\ Rate = \frac{N_{Delivered}}{N_{Attempted}} \times 100 \quad (11)$$

This demonstrates that the Haversine-based distance computation provides accurate geofencing and reliable event detection even in dense urban conditions.

C. AI-Assisted Grievance Classification

The grievance management system employed a Natural Language Processing (NLP)-based model hosted on Firebase to automatically classify textual complaints. The dataset included 1,500 complaint entries across five classes: *missed collection*, *overflow*, *delayed pickup*, *unclean area*, and *others*.

The classification metrics were calculated using the same precision–recall framework as in Section VI-A. The NLP model achieved an accuracy of 90.1% and an F1-score of 0.89. The average inference time for each complaint was 0.7 seconds, corresponding to a throughput of:

$$Throughput = \frac{1}{0.7} = 1.43 \text{ requests/sec} \quad (12)$$

Automation led to a 75% reduction in manual complaint sorting time, yielding a 4× improvement in operational speed:

$$Speedup = \frac{1}{1-0.75} = 4.0 \times \quad (13)$$

This improved prioritization of high-impact complaints and ensured faster routing to municipal field teams.

Module	Accuracy / Success Rate	Latency / Inference Time	Derived Metric
AI Waste Segregation	85.2%	95 ms	F1 = 0.85; Throughput = 10.53 img/s
Dynamic Geofencing	96.4%	1.8 s	Alert reliability 22%
NLP Grievance Classifier	90.1%	0.7 s	F1 = 0.89; Speedup = 4.0×

Table 2: Consolidated System Performance

The combined findings demonstrate that ReGenX effectively integrates multiple intelligent subsystems into a single operational framework. The system achieved high notification reliability, accurate AI inference performance, and efficient complaint management. The results validate ReGenX as a scalable, citizen-centric, and data-driven solution for urban waste management.

VI. CONCLUSION AND FUTURE WORK

ReGenX presents a powerful, software-based model for urban waste management and inspires new directions in urban waste accountability and public service. The ReGenX platform shows that mobile technology and on-device AI represent a cost-effective diversion from the high infrastructure investments associated with the conventional IoT-based models. When compiled, dynamic geofencing notifications, AI-enabled grievance handling, and waste sorting functionality within one mobile platform effectively creates a seamless linkage between municipal waste management activities and civic accountability. Synthesis of results: The results of the evaluation confirm the effectiveness of this approach. The dynamic geofencing notifications experienced a 96.4% success rate and did so with quick notification visibility to the residents - addressing their long-standing concern of not knowing when waste collection vehicles would be in their area. The Natural Language Processing (NLP) software for grievance redress achieved an impressive classification accuracy of 90.1% and 4x faster operational speeds outwardly handling

the grievance sorting that had previously been a manual process. Finally, the lightweight, on-device TensorFlow Lite waste sorting model achieved classification accuracy at 85.2% and an average inference time of just 95ms which speaks to the technical and economic feasibility of implementing low-resource AI applications in mobile situations. Altogether, these findings provide evidence that ReGenX has the potential to improve service effectiveness, transparency and citizen engagement. Limitations and Future Directions: This prototype has two main limitations. First, it operates in a vacuum of existing waste infrastructure (there are no real-time updates of waste levels from IoT-enabled smart bins). Second, the system needs a wider stress test and deployment in different, large urban settings to assess the scalability and robustness of the solution with larger user loads. To mitigate these limitations, future work will take several directions. First, we will integrate IoT technologies with low-energy, low-cost sensors to enable real-time detection of bin waste levels and collections, which will lead to predictive modelling for demand. Second, advanced AI capabilities will be developed, such as multi-class visual models for separation and accurate waste destination selection, and employing genetic algorithms or reinforcement learning to optimize intelligent route planning to minimize fuel costs. Finally, governance and policy integration will be a focal point for engaging enterprise deployment with municipal regulations, and protocols will be established for privacy and transparency in complaints and waste data. As these developments advance, ReGenX can become a fully autonomous, omnichannel waste management model, a scalable and replicable solution for future urban sustainability practices.

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